

# Network Infrastructure Day 5 Summary

Friday, October 9<sup>th</sup>, 2026

## Overview

Key participants: QA Trainer Ninad Rampure leads and instructs throughout, with active student contributors including Arley Gibbons, Fabian, Kai Standley, Blake, Rion Schofield, Liam, Thomas Kent, Krystian, Ali and others. The session begins with Packet Tracer lab troubleshooting (RIP and EIGRP), where Ninad diagnoses routing and interface misconfigurations, guides static IP fixes and RIP/EIGRP network additions, and instructs students to verify routes with `show ip route`. He coordinates lab time, posts replab corrections, and asks students to report completion in chat.

After labs, Ninad delivers a cloud fundamentals and virtualization module covering service models (IaaS/PaaS/SaaS/X-as-a-Service), deployment models (public/private/hybrid/community/sovereign), multi-cloud strategies (Azure Arc), and cost/security considerations including egress charges and region selection. He follows with virtualization topics (hypervisors, VMs vs containers) and transitions to structured troubleshooting: OSI-based approaches, common tools (ping, traceroute, Wireshark, Nmap, SNMP monitors, Azure Monitor/Advisor) and complementary fault-diagnostic techniques (five whys, fishbone, fault tree, chronological analysis).

The latter portion focuses on switching and LAN troubleshooting: practical checks for port LEDs and cabling, configuration pitfalls (trunk vs access, all-ports-as-trunks), VLAN verification via CLI or centralized management, inter-VLAN routing, and STP roles to prevent loops. IP addressing and DHCP planning are reinforced, including APIPA causes, lease-duration impacts, and the critical role of correct default gateway and subnet masks. Ninad also clarifies VLSM/subnetting activity questions raised by Fabian and Arley, confirms corrections (including counting point-to-point subnets), and explains how to allocate multiple subnets (e.g., /21s) inside a larger /16.

Major outcomes and commitments: Ninad posts replab fixes to chat and GitHub and will update the subnetting activity answers; students commit to applying IP/interface fixes in labs and completing Packet Tracer questionnaires and reflection templates for certification. Resources and next steps: Ninad points learners to GitHub for lab files and lesson summaries, recommends downloading Packet Tracer from Cisco, and encourages optional further study (CompTIA Network+, Microsoft Azure/365) and following up with DLCs for exam vouchers or deadline extensions. Remaining concerns include ensuring correct interface/subnet configuration to avoid routing/EIGRP issues, enforcing cloud cost governance, and maintaining VLAN/STP correctness to prevent switching loops.

# Detailed Summary

## RIP lab troubleshooting

Trainer Ninad identifies routing mismatches causing ping failures in the RIP lab and explains fixes students should apply. He traces the issue to inconsistent IP/subnet assignments between DHCP and RIP configurations and demonstrates how adding the correct network to RIP and statically assigning missing PC IPs resolves connectivity. The lab is now passing pings after the corrections and Ninad confirms how to verify routes.

- QA Trainer Ninad Rampure explains the ping failures are due to one wrong IP/subnet and inconsistent DHCP versus RIP configurations.
- QA Trainer Ninad Rampure instructs Ali to statically configure PC2 as 2.1 with gateway 2.254 to make the device reachable for pings.
- QA Trainer Ninad Rampure recommends adding the 30.0.0.0 network (or 30.10.0.0 in class-C style) to RIP to include the DHCP-assigned subnet in the routing domain.
- QA Trainer Ninad Rampure demonstrates that the first ping to a newly learned route can fail because ARP occurs, and subsequent pings succeed.
- QA Trainer Ninad Rampure advises students to run `show ip route` to verify the RIP-populated routes (marked with R).

## EIGRP lab interface checks

Students encounter EIGRP network command and discontinuous mask errors; Ninad diagnoses missing interface IP configurations on several routers. He advises verifying directly connected (C) routes in `show ip route` and ensuring each router interface is configured and up before enabling EIGRP. Fixing the missing interface configurations allows EIGRP to form neighbor relationships and populate routes.

- Kai Standley reports EIGRP invalid address/discontinuous mask errors when entering the network command for a subnet.
- QA Trainer Ninad Rampure identifies that one interface on Atlanta and Chicago routers is not configured and instructs students to configure the missing interface addresses.
- QA Trainer Ninad Rampure tells students to confirm directly connected routes (C entries) with `show ip route` before troubleshooting EIGRP.
- QA Trainer Ninad Rampure confirms New York router has both interfaces configured and thus accepts the EIGRP network command without error.
- QA Trainer Ninad Rampure instructs everyone to ensure interfaces are administratively up and configured to avoid inconsistencies across the topology.

## Lab completion coordination

Ninad monitors lab progress, extends lab time, posts corrective steps in chat, and instructs students on next steps once labs finish. He asks students to notify him in chat or by raising hands when they complete labs so the class can pause for a break and move to the next module. He also notes these are the final Packet Tracer labs.

- QA Trainer Ninad Rampure extends lab time to allow students to complete EIGRP and posts replab changes into the chat for reference.
- QA Trainer Ninad Rampure requests students to indicate completion in chat or raise hand so he can plan the start of day five.
- QA Trainer Ninad Rampure states these are the last Packet Tracer labs and asks students to finish them before moving to the next module.

## Cloud computing introduction

Ninad leads an interactive session on cloud concepts, asking students to define cloud computing and distinguish it from on-prem infrastructure. He frames cloud as someone else's infrastructure accessed as services (SaaS, PaaS, IaaS), emphasizes subscription/OPEX models, and introduces virtualization and containerization ideas relevant to platform services.

- QA Trainer Ninad Rampure defines cloud computing as virtualized, internet-available resources hosted on someone else's infrastructure and emphasizes the OPEX pay-as-you-go model.
- Rion Schofield describes cloud as remote storage and virtualized file access compared to local on-prem storage.
- QA Trainer Ninad Rampure explains SaaS/PaaS/IaaS distinctions and notes that services like Exchange/Office 365 and OneDrive are examples of SaaS and storage services.

## Cloud service examples and roles

The group discusses concrete cloud examples (OneDrive, SharePoint, Exchange, Netflix) and how different cloud service models map to real products. Ninad highlights containerization for PaaS, describes SharePoint and Office 365 as collaborative SaaS offerings, and explains how companies can combine cloud models in hybrid deployments.

- Arley Gibbons identifies OneDrive and SharePoint as cloud services used daily.
- Blake points out running virtual machines and hardware abstraction as examples of IaaS usage.

- QA Trainer Ninad Rampure explains Netflix uses cloud provider infrastructure and builds platform/services on top as a real-world example spanning IaaS/PaaS/SaaS.

## **Elasticity and scalability**

Ninad defines cloud elasticity versus on-prem scalability, explaining that cloud allows both scale-out (horizontal) and scale-up (vertical) on demand and enables pay-for-what-you-use. He outlines horizontal scaling via VM scale sets and vertical scaling via PaaS/container resource adjustments, and explains how elasticity reduces capital expenditure and supports fluctuating demand.

- Rion Schofield equates elasticity with the ability to scale resources up or down based on demand.
- QA Trainer Ninad Rampure contrasts on-prem capex-driven scaling with cloud elasticity that enables scaling in and out and pay-as-you-go OPEX.
- QA Trainer Ninad Rampure describes horizontal scaling (scale sets adding instances) and vertical scaling (adding CPU/memory in PaaS/container scenarios).

## **Cloud benefits for business continuity**

Ninad explains how cloud enhances disaster recovery and business continuity through regional replication, availability zones, geo-redundant backups, and provider-managed redundancy. He stresses choosing regional placement for performance and explains how paired regions and multiple availability zones reduce outage risk.

- QA Trainer Ninad Rampure explains that cloud providers use availability zones and regional pairing to replicate data and support disaster recovery.
- QA Trainer Ninad Rampure highlights geo-redundant and zone-redundant storage options as choices that affect backup behavior and recovery.
- QA Trainer Ninad Rampure states selecting the appropriate region (e.g., UK South vs. UK West) improves user experience and supports regulatory/latency needs.

## **Cloud for remote work and endpoint management**

The class reviews cloud advantages for remote work, mentioning cloud PCs, VDI, Microsoft 365, Intune, Autopilot, and centralized endpoint management. Ninad emphasizes faster provisioning, remote deployment, unified device management, and security monitoring tools that support hybrid work models.

- Arley Gibbons notes cloud enables remote access to on-site machines and resources during remote sessions.
- QA Trainer Ninad Rampure highlights cloud PC and VDI as ways to provide persistent remote desktops accessible from anywhere.
- Blake explains Intune allows remote deployment of software and updates across devices, facilitating fast provisioning and management.

## **Cloud security and cost considerations**

Discussion covers security risks (public network exposure, provider targeting) and cost models (pay-per-resource, egress charges). Ninad outlines VPN/ExpressRoute for private connectivity, Defender suites for monitoring, and cautions about ongoing operational costs and data egress fees that can raise monthly bills.

- Participants identify public network exposure and provider targeting by attackers as primary cloud security concerns.
- QA Trainer Ninad Rampure recommends VPN gateways or ExpressRoute for private or encrypted connectivity to mitigate exposure.
- Liam and QA Trainer Ninad Rampure point out cloud operating costs can add up and that egress and continuous VM uptime drive significant charges if not planned.

## **Sovereign and multi-cloud concerns**

Ninad discusses data sovereignty, regional legislation, and vendor lock-in as privacy and availability concerns that influence cloud strategy. He explains sovereign clouds and community clouds as region- or sector-specific deployments and recommends multi-cloud and Azure Arc as approaches to mitigate vendor lock-in and provider outages. The conversation also covers cost control measures and governance to avoid runaway cloud spending.

- QA Trainer Ninad Rampure explains China's sovereign cloud policies and how region-specific laws force data residency decisions.
- QA Trainer Ninad Rampure attributes regulatory choices like GDPR and national policies to where organizations must keep their data, citing TikTok and other examples.
- QA Trainer Ninad Rampure recommends multi-cloud (managed via Azure Arc) as a strategy to reduce vendor lock-in and limit impact of a single provider outage.
- QA Trainer Ninad Rampure advises setting auto-scaling limits and scavenging policies to shut down unused developer VMs to control cloud costs.
- QA Trainer Ninad Rampure warns that ancillary services (firewalls, VPNs, public IPs, cross-region data flows) add to total cost of ownership and must be included in estimates.

## Cloud deployment models

Ninad explains public, private, hybrid, and community cloud models and gives guidance on selecting a model based on scalability, accessibility, security, and cost. He emphasizes hybrid for combining cloud scalability with on-prem data control and shows how community or sovereign clouds serve sector- or government-specific needs.

- Blake defines public cloud as multi-tenant and private cloud as organization-specific, and Ninad affirms this distinction.
- QA Trainer Ninad Rampure describes community clouds as shared-services clouds for industry groups (e.g., apprenticeship providers or healthcare) and gives gov.uk as an example.
- QA Trainer Ninad Rampure explains hybrid cloud provides the best of both worlds by running apps in public cloud while keeping sensitive data on-premise for control and compliance.

## X-as-a-Service and service models

The class covers IaaS, PaaS, SaaS and the broader X-as-a-Service ecosystem, illustrating each model with practical examples. Ninad clarifies responsibilities (e.g., IaaS users manage VMs/OS, PaaS users manage apps only, SaaS users consume applications) and highlights niche services such as Desktop-as-a-Service and Bare-metal-as-a-Service for DR use cases.

- QA Trainer Ninad Rampure defines IaaS as virtualized hardware where users manage VMs and OS, PaaS as provider-managed OS/platform focusing user responsibility on apps, and SaaS as fully managed applications.
- Blake and Liam give examples including Microsoft 365 and Google apps as SaaS and VMs as IaaS.
- QA Trainer Ninad Rampure mentions examples of X-as-a-Service like storage-as-a-service, communication-as-a-service, desktop-as-a-service, and a Linux bare-metal disaster recovery service.

## Virtualization fundamentals

Ninad teaches virtualization concepts, explaining hypervisors, resource pooling, and types of virtualization for servers, storage, and networks. He stresses how virtualization improves hardware utilization and simplifies management but introduces single points of failure unless clustered or replicated for high availability.

- QA Trainer Ninad Rampure explains virtualization as running many virtual machines on one physical host to improve hardware utilization.
- QA Trainer Ninad Rampure lists server, storage, and network virtualization examples including SAN, virtual switches, and VLANs.

- QA Trainer Ninad Rampure cautions that virtualization centralizes management and can create single points of failure without clustering and replication.

## **Hypervisor types and use cases**

Ninad differentiates Type-1 (bare-metal) and Type-2 (hosted) hypervisors, describing performance, management, and lab/test use cases. He explains Type-1 (e.g., ESXi, Hyper-V Server) is suited to production for performance and control, while Type-2 is convenient for development and local labs and can be used to manage Type-1 instances remotely.

- QA Trainer Ninad Rampure defines Type-1 hypervisors as bare-metal (e.g., ESXi, Hyper-V Server) offering higher performance and suitability for production.
- QA Trainer Ninad Rampure describes Type-2 hypervisors as hosted on a general OS and useful for development, testing, and lab environments.
- QA Trainer Ninad Rampure notes administrators can use Type-2 tools to manage Type-1 hosts remotely through management applications like vSphere.

## **VMs vs containers**

Ninad contrasts virtual machines and containers: VMs provide separate guest OS instances while containers share a host OS and package application dependencies. He explains containers are more efficient for single-purpose, multi-user application delivery (PaaS scenarios), while VMs are preferable for isolated, full-OS workloads and certain high-resource use cases.

- QA Trainer Ninad Rampure explains VMs include full guest operating systems while containers share the host OS and package app dependencies.
- QA Trainer Ninad Rampure highlights containers and Docker as cost-effective for multi-user app access and PaaS delivery, reducing per-user VM overhead.
- QA Trainer Ninad Rampure states VMs better fit scenarios that require full OS isolation, whereas containers fit shared-platform, scalable app deployments.

## **Troubleshooting concepts and tools**

After lunch, Ninad transitions to troubleshooting methodology, emphasizing the business impact of unresolved incidents and the need for structured diagnostics. He lists common tools (ping, nslookup, traceroute, Wireshark, Nmap, SNMP-based monitors, Fluke cable testers, PRTG, SolarWinds, Azure Advisor/Monitor) and recommends using monitoring data to identify root causes before outages escalate.

- QA Trainer Ninad Rampure frames troubleshooting as essential to maintain SLAs and prevent customer-impacting downtime.
- Rion Schofield highlights that patch fixes can mask underlying causes without proper root-cause analysis.
- QA Trainer Ninad Rampure lists troubleshooting tools including ping, nslookup, traceroute, Wireshark, tcpdump, Nmap, SNMP monitors, PRTG, SolarWinds, and Fluke cable testers.
- QA Trainer Ninad Rampure recommends cloud-native tools (Azure Advisor and Azure Monitor) for diagnosing cloud resource issues.

## **Troubleshooting methodologies and OSI use**

Ninad explains structured troubleshooting approaches—top-down, bottom-up, and divide-and-conquer—using OSI layering to isolate faults. He demonstrates practical steps: verify physical connectivity and IP, test gateway reachability, then move up to application-layer checks (DNS, firewall, app config) to pinpoint whether issues are link, routing, or application related.

- QA Trainer Ninad Rampure advises starting with physical/connectivity checks, then verifying IP and gateway reachability before moving up the OSI layers.
- Blake summarizes the approach as checking from lower layers upward to locate the failing layer.
- QA Trainer Ninad Rampure gives a real example where firewall rules blocked YouTube by name while IP connectivity remained, illustrating DNS vs application filtering distinctions.
- QA Trainer Ninad Rampure frames troubleshooting as divide-and-conquer: isolate whether the problem is hardware, network, or application then act accordingly.
- QA Trainer Ninad Rampure introduces additional fault-diagnostic frameworks and techniques such as five whys, fishbone (Ishikawa), fault tree analysis, value analysis, and chronological analysis as complementing OSI-based troubleshooting.

## **Switch faults and port issues**

Ninad covers common switch-level faults and practical checks to identify port and hardware failures. He emphasizes visual inspection of link LEDs, consistent cabling and patch-panel standards, and monitoring resource utilization to detect abnormal switch behavior that can indicate misconfiguration or hardware failure.

- QA Trainer Ninad Rampure explains LED states (blinking green, steady green, orange, red) as quick indicators of link status and performance.

- QA Trainer Ninad Rampure warns that unmanaged changes to patch-panels and cabling cause mapping chaos and recommends standardized cable management.
- QA Trainer Ninad Rampure recommends benchmarking switch resource utilization and investigating devices running consistently at unusually high CPU or memory.

## Switch configuration, VLANs and STP

The class reviews switch configuration best practices, VLAN/trunk/access port distinctions, inter-VLAN routing, and Spanning Tree Protocol to prevent switching loops. Ninad demonstrates STP roles (root bridge, root port, blocked port), identifies common misconfigurations (all ports set to trunk, wrong VLAN assignments), and suggests CLI and centralized tools to verify VLAN state across devices.

- QA Trainer Ninad Rampure explains STP elects a root bridge and places one port per switch in a blocked state to prevent loops and recommends verifying STP state in Packet Tracer or on devices.
- QA Trainer Ninad Rampure warns that misconfigured trunk/access ports or all-ports-as-trunks can expose VLANs incorrectly and cause connectivity issues.
- QA Trainer Ninad Rampure suggests using show vlan and other vendor CLI commands or centralized management (Meraki, Aruba, HPE) to verify VLAN assignments across switches.
- QA Trainer Ninad Rampure highlights inter-VLAN routing via layer-3 switch SVI/subinterfaces and cautions missing VLANS in routing configuration isolates traffic.

## IP addressing and DHCP planning

Ninad reiterates IP addressing fundamentals, subnet selection, DHCP lease sizing and consequences of wrong default gateways or subnet masks. He explains APIPA symptoms, lease-duration impacts for wireless vs fixed devices, and how incorrect gateway or mask configuration prevents inter-subnet communication and causes apparent outages.

- QA Trainer Ninad Rampure explains that an incorrect default gateway or subnet mask stops hosts from leaving their subnet and prevents routing to other networks.
- QA Trainer Ninad Rampure identifies APIPA addresses as signs of DHCP issues or exhausted pools and recommends appropriate lease durations for wireless versus static hosts.
- QA Trainer Ninad Rampure advises planning subnet sizes (CIDR/slash notation) to match expected host counts to avoid wasted or insufficient address space.

## Routing fundamentals: static vs dynamic

Ninad contrasts static routing and dynamic protocols, describing appropriate use cases and failure modes. He outlines how static routes suit small stable networks while EIGRP/OSPF handle complex topologies, and stresses verifying show ip route outputs to confirm topology-wide route propagation when troubleshooting routing issues.

- QA Trainer Ninad Rampure explains static routes require manual entries on every router and are suitable for small, stable networks.
- QA Trainer Ninad Rampure contrasts dynamic protocols (EIGRP, OSPF, RIP) by capability and use-case, noting RIP is simple but limited and EIGRP/OSPF are richer for complex topologies.
- QA Trainer Ninad Rampure recommends using show ip route to check which networks each router knows and to identify interface or protocol misconfigurations when routes are missing.

## Course wrap-up and resources

Ninad closes the module by summarizing covered topics (OSI, switching, VLANs, IP addressing, routing, cloud, virtualization, troubleshooting) and provides resources and next steps. He directs students to GitHub for lab files and the lesson summary, reminds them to complete Packet Tracer questionnaires and reflection templates, and suggests further certifications and study options like CompTIA Network+ and Microsoft Azure/365.

- QA Trainer Ninad Rampure summarizes the module coverage and points learners to GitHub for lab files and the lesson summary.
- QA Trainer Ninad Rampure asks students to complete packet tracer lab questionnaires and reflection templates to qualify for certification and suggests contacting the DLC for deadline extensions or exam vouchers.
- QA Trainer Ninad Rampure recommends pursuing further certifications (CompTIA Network+, Microsoft Azure/365) and degree apprenticeship options for long-term career development.

## VLSM and subnetting clarifications

Ninad answers detailed student questions on VLSM and the subnetting activity, clarifying how to count point-to-point subnets and how to apply varying masks across portions of a classful network. He confirms activity corrections (initial count updated to include router interfaces) and explains reusing larger subnets (e.g., /21 within a /16) to satisfy different host requirements across the design.

- Fabian raises a subnet-count discrepancy and QA Trainer Ninad Rampure confirms point-to-point router interfaces increase the total subnet count and updates the activity to nine subnets.
- Fabian questions borrowing bits between sequential VLSM subnets and QA Trainer Ninad Rampure explains you can allocate multiple /21 subnets within the original /16 block to satisfy separate host requirements.
- QA Trainer Ninad Rampure recommends using subnet tables and incremental ranges when planning VLSM and notes architects typically perform this planning.